

1 Algorithms Introduction

- Insertion Sort Example

Input: 503 87 512 61 908 170 897

`/* The brackets signify the sorted array */`

```
[503] 87 512 61 908 170 897
[87 503] 512 61 908 170 897
[87 503 512] 61 908 170 897
[61 87 503 512] 908 170 897
[61 87 503 512 908] 170 897
[61 87 170 503 512 908] 897
[61 87 170 503 512 908] 897
[61 87 170 503 512 897 908]
```

- Do the take home examples.

2 Asymptotic Analysis

- Big-O

$$O(g(n)) = \{f(n) : \text{there exists positive constants } c \text{ and } n_0 \text{ such that } 0 \leq f(n) \leq cg(n) \forall n \geq n_0\}$$

(1)